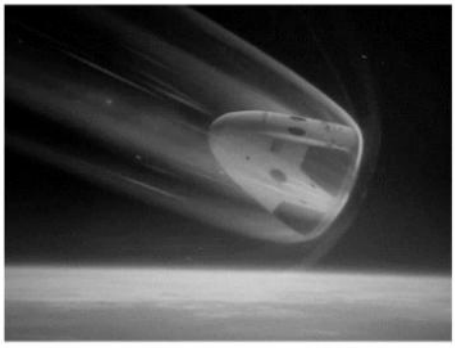
P-set 1

Tuesday, February 3, 2026 5:03 PM

1. (30 pt) Consider a spacecraft traveling in the space *outside* the atmosphere, discuss the dominant heat transfer mode across the spacecraft and briefly explain. After completing the space exploration mission, this spacecraft returns to earth by traveling *through* the atmosphere. Please discuss the dominant heat transfer mode across the spacecraft and briefly explain.

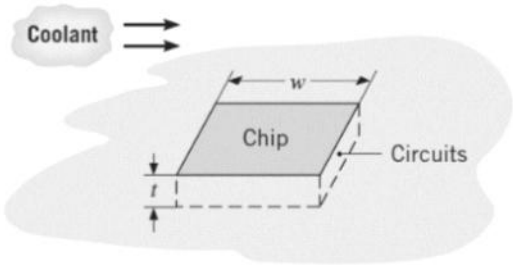


Note: you do not need to follow the recommended format for this problem, since this is a short-answer problem.

For a spacecraft travelling outside of the atmosphere, the primary heat transfer mode transferring heat to and from the spacecraft would be radiation. In space there is essentially no fluid or other matter around the spacecraft that would allow conduction or convection to take place, so radiation from the spacecraft and other sources (such as the sun) would be the only significant form of heat transfer.

For spacecraft traveling through the atmosphere, the primary source of heat transfer would be convection, along with radiation when the spacecraft becomes extremely hot. We model heat transfer through convection using Newton's law of cooling, and according to this law, heat transfer through convection is directly related to the coefficient "h". When moving through the atmosphere at a high velocity, this coefficient will be very large making convection the dominant heat transfer mode.

2. (30 pt) A square silicon chip (k = 150 W/(m·K)) is of width w = 5 mm on a side and of thickness t = 1 mm. The chip is mounted in a substrate such that its side and back surface are thermally insulated. The front surface is exposed to a coolant with convective flows. The chip dissipates 9 W energy under steady-state operation. Assume that thermal radiation is negligible in this problem. The coolant temperature is T_∞ = 20 °C.



(a) What is the steady-state temperature difference between back and front surfaces of the chip?

(b) If the front surface temperature of the chip cannot be above 80 °C, calculate the minimum heat transfer coefficient required for the coolant flow. Compare your calculation with heat transfer coefficients shown in Table 1.1 of the textbook and discuss what convection processes can be used to cool this chip.

Known

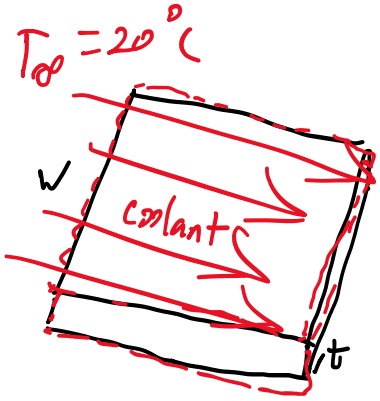
- $k = 150 \frac{W}{m \cdot K}$
- $W = 5mm$
- $t = 1mm$
- $T_\infty = 20°C$
- $Q = 9W$

b) $T_{front} \leq 80°C$

Find: a) ΔT_{ss} between back and front of chip

b) Minimum heat transfer coeff for coolant flow if T_{front} must be less than 80°C

Schematic



Assumptions:

- Thermal rad is negligible
- 1D heat conduction
- Side and back insulated perfectly
- Constant thermal conductivity
- Steady state

Analysis

Since all qw must leave from the front surface, heat flux can be found via $q''_{front} = \frac{Q}{A} = \frac{q}{2.5\times10^{-5}} = 3.6\times10^{5}\ \frac{W}{m^2}$

$q''_{back} = 0$ due to insulation

q''_{avg} will be the average of q''_{front} and q''_{back}

$q''_{avg} = \frac{1}{2} q''_{front}$

Now Applying Fourier's law:

$$q''_{avg} = -k\frac{dT}{dx} = 150\ \frac{T_{back} - T_{front}}{t}$$

$$T_{back} - T_{front} = \frac{3.6\times10^{5} \cdot 1\times10^{-3}}{2 \cdot 150} = \boxed{1.2\,^{\circ}C}$$

Due to insulation all q_W of power dissipation occur on the front surface due to convection, and therefore

$$Q = q_W = hA(T_s - T_\infty)$$

$$h = \frac{Q}{A(T_s - T_\infty)} = \frac{q_W}{(2.5\times10^{-5})(80-20)} = \boxed{6\times10^{3}\ \frac{W}{m^2K}}$$

Table 1.1 Comparison

- To reach a value of $h = 6000\ W/m^2K$ you would most likely use forced liquid convection which has a coefficient of $100 - 20000\ \frac{W}{m^2K}$

3. (40 pt) The emissivity of a roof material is $\varepsilon = 0.18$. The absorptivity of this material for solar irradiation is $\alpha_s = 0.76$.

(a) Consider a day of solar irradiation $G_s = 750$ W/m^2, ambient temperature $T_\infty = 300$ K, surrounding temperature $T_{surr} = 300$ K, and heat transfer coefficient of air flow $h = 10$ W/(m^2· K), calculate the temperature of the roof. Assume the bottom surface of the roof is thermally insulated. Assume the roof can be treated as a grey body to the irradiation of surrounding environment. Would the neighboring cat be comfortable walking on the roof constructed by this material?

(b) According to your calculation, discuss the dominant heat transfer mode in (a).

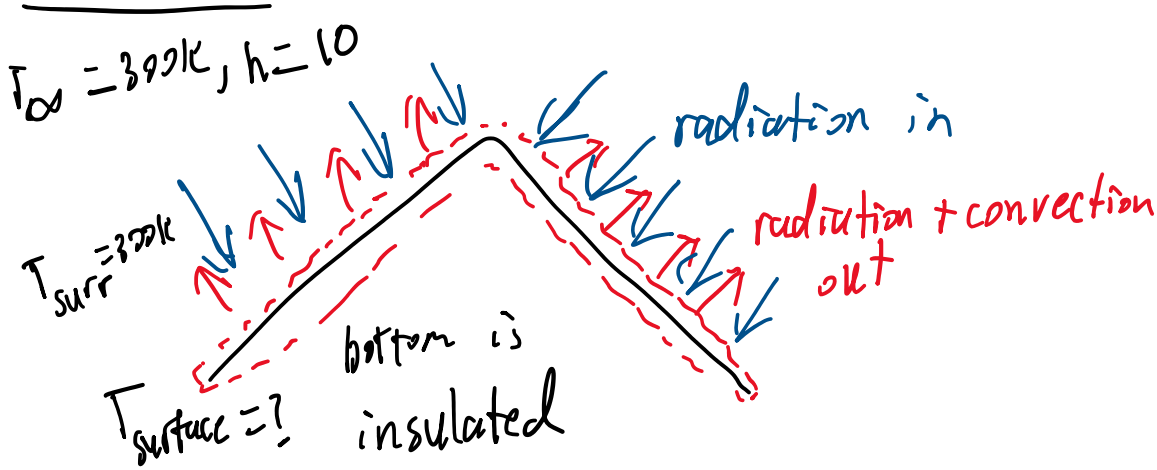
Known:

- $\varepsilon = 0.18$, $\alpha_s = 0.76$, $G_s = 750 \frac{W}{m^2}$, $h = 10 \frac{W}{m^2 K}$, $T_\infty = 300K$, $T_{sur} = 300K$, $\sigma = 5.67 \times 10^{-8} \frac{W}{m^2 K^4}$
- Bottom is insulated
- Roof is grey body

Find: a) Temperature of the roof, would a cat be comfortable

b) dominant heat transfer mode

Schematic:



Assumptions:

- Roof temp is same throughout
- Bottom perfectly insulated
- Uniform solar-rad
- constant h
- Steady state conditions

Analysis

3 modes of heat exchange take place:

1. absorbing solar radiation, given by

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$$q''_{solar} = \alpha_s G_s$$

2. heat loss due to convection, given by

$$q''_{conv} = h(T - T_\infty)$$

3. heat radiation loss, given by

$$q''_{rad,out} = \varepsilon\sigma(T^4 - T_{surr}^4)$$

At steady state the entire heat transfer can be represented by:

$$q''_{solar} = q''_{conv} + q''_{rad,out}$$

$$\downarrow$$

$$\alpha_s G_s = h(T - T_\infty) + \varepsilon\sigma(T^4 - T_{surr}^4)$$

After plugging in known values:

$$T \approx 350K = \boxed{77°C}$$

<u>The cat would not be comfortable</u>

b) Solar heat: $\alpha_s G_s = 570\ W/m^2$

Convection: $q''_{conv} = h(T - T_\infty) = 500\ W/m^2$

rad out: $q''_{rad} = \varepsilon\sigma(T^4 - T_{surr}^4) = 70\ \frac{W}{m^2}$

The largest mode is the influx of heat due to solar radiation, but the loss due to convection balances out most of it.